

MEMVERSE — Zero-Trust Privacy Gateway for AI Memory

Solving the Personalization vs. Privacy Paradox for Text, Documents & Multimodal AI

🔗 Project Description (2-3 Line Summary):

MEMVERSE is a client-governed zero-trust privacy gateway that empowers AI assistants to deliver highly personalized responses without exposing raw personal identifiers to external LLMs. Through an automated 12-stage security pipeline, dynamic policy transformation (names to roles, exact numbers to bands, locations to regions), cryptographic Memory Passports, and tamper-evident SHA-256 ledger receipts, MEMVERSE guarantees mathematical privacy for text, PDF documents, and biometric images.

👥 Team Information

Role	Name	Institution & Details
Team Lead & Developer	Satvik Kesarwani	Indian Institute of Information Technology (IIIT), Pune B.Tech — Computer Science & Engineering (MIS: 112315166)
Team Name	Team MEMVERSE (Zero-Trust AI Privacy Initiative)	

🔗 Project Links & Live Deployments

🌐 Live Web Application (Vercel)

Interactive React/Vite app with ChatView, Identity Vault, and live Trace Radar:

<https://memverse-satvikkesarwanis-projects.vercel.app>

⚙️ Gateway API Service (Render)

FastAPI zero-trust gateway with live NVIDIA NIM model integration:

<https://memverse-api.onrender.com/api/status>

💻 Source Code Repository (GitHub)

Full monorepo including frontend, backend, test suite & Playwright E2E:

<https://github.com/satvikkesarwani/memverse>

📁 Memverse Project Details (Drive)

Submission folder with video demo, architectural assets & documentation:

https://drive.google.com/drive/folders/1U03jMmCsiOP9CiwQckT6S5NyeJYVo_f9?usp=sharing